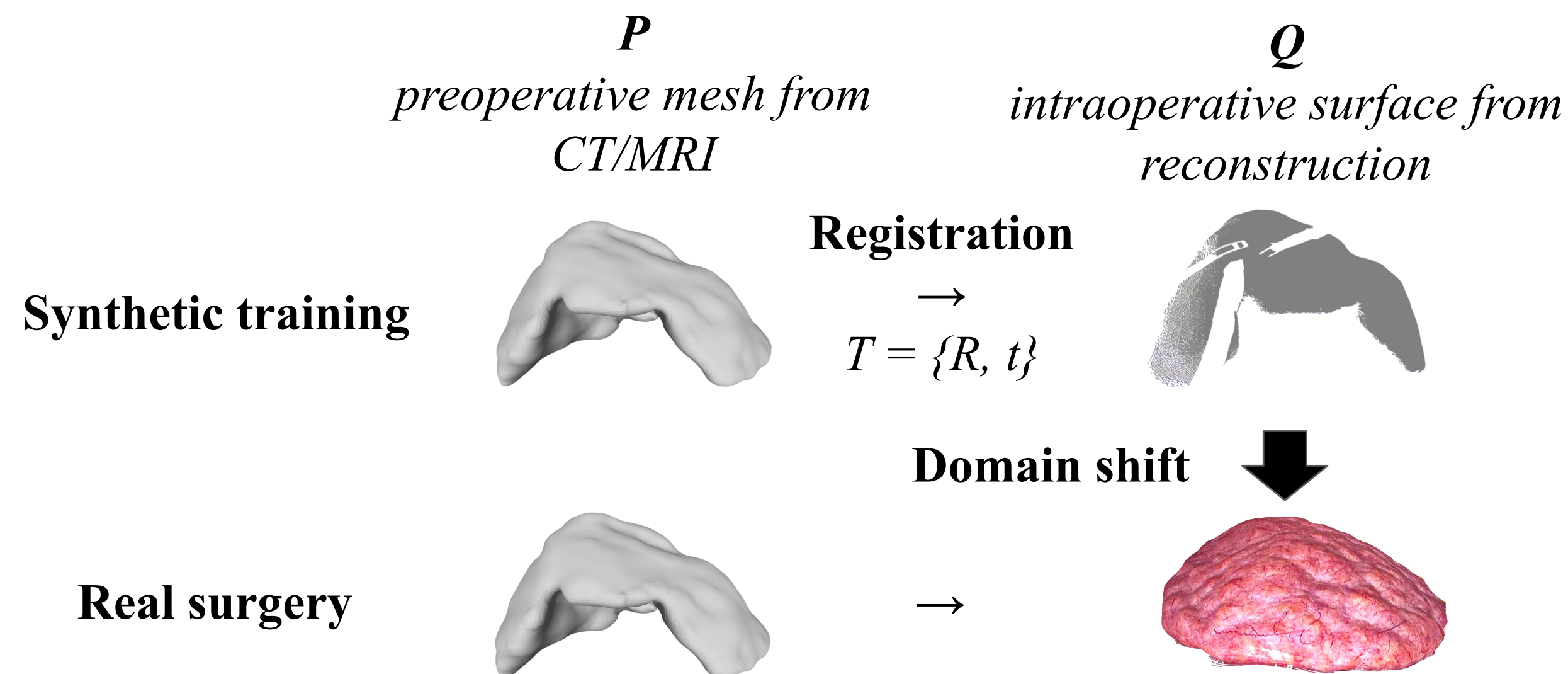


Introduction

Domain shift in laparoscopic registration

- **Rigid registration** : Align P and Q by estimating a transformation $T = \{R, t\}$.
- **Problem** : Models are **trained on synthetic pairs** for the ground truth T and **tested on real ones**, introducing a domain shift.



- **Test Time Adaptation (TTA)**: adapts the model at inference, using incoming unlabeled target data, without access to source data at test time.

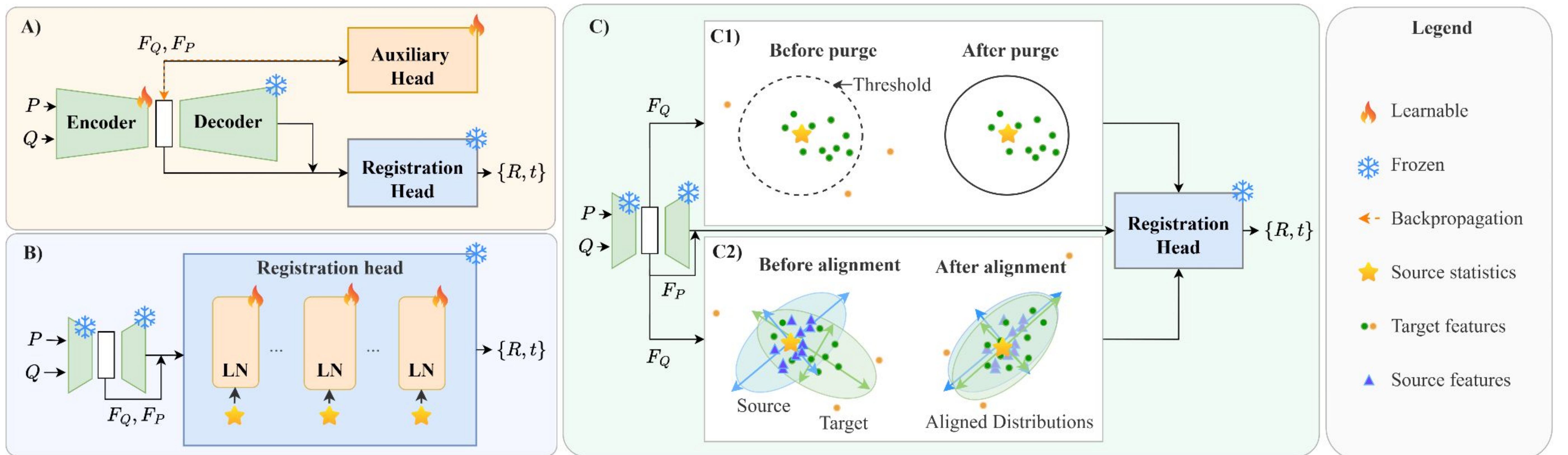
Why is TTA challenging for registration ?

- Most TTA methods are designed for classification/segmentation, and rely on mechanisms that do not directly transfer to registration.
- TTA methods adapt a single input, whereas registration adapts a pair (P, Q) with an asymmetric shift, mainly affecting Q .

Main contributions

- (1) **Analysis and modification of four TTA methods from three TTA families**:
 - Modified to account for the asymmetric domain shift between P and Q .
 - Replace Purge-Gate's classification entropy criterion with an inlier ratio (IR) mechanism.
- (2) **Comparative evaluation of the adapted methods under domain shifts**.
 - Target data: real data (real shift)/ synthetic data plus corruptions (synthetic shift), 8 corruptions with 5 levels of severity.
 - On P2P and P2ILReg.

Methodologies



A) Model adaptation - Point-TTA [12]

- Update the shared encoder using a reconstruction task and backpropagation.

B) Normalization adaptation - LN [40]

- Adapt *LayerNorm* (LN) post-affine parameters using source and target statistics.

Input adaptation

- Adaptation is applied to the input features before registration

C1) Purge-Gate [39] : remove target features that diverge the most from the source distribution.

C2) PEA [20]: align source and target statistics at one level. PEA-ML aligns statistics at multiple levels.

Results

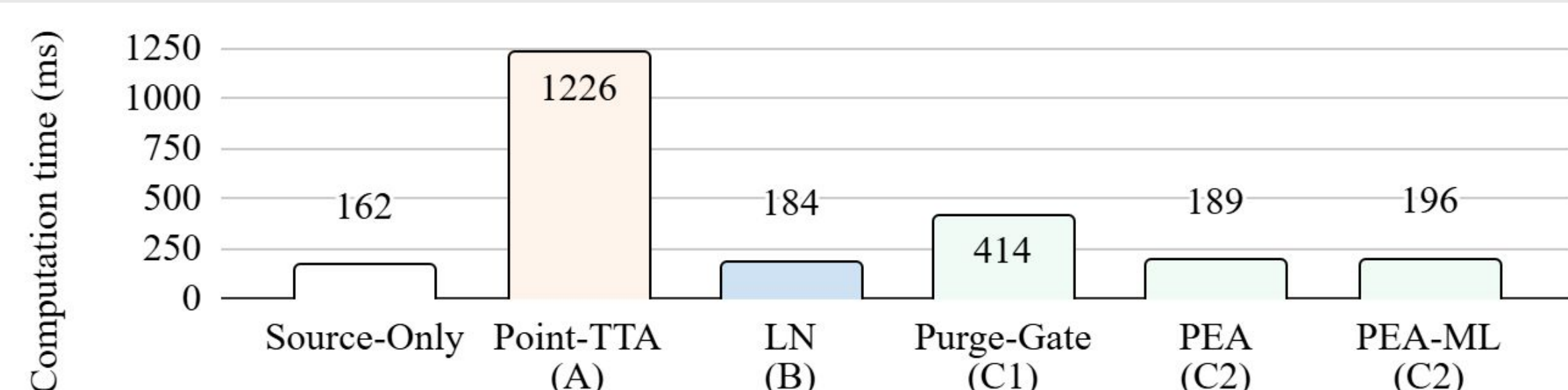
Main results

	Method	P2P-Syn Syn → Corrupted syn	P2ILReg-Syn Syn → Corrupted syn	P2ILReg-Real Syn → Real	
		RMSE (mm)↓	RMSE (mm)↓	CD (mm)↓	Dice ↑
EP	Source only	15.47	38.70	4.26	71.87
	Point-TTA (A)	14.75	-	-	-
	LN (B)	10.06	47.31	5.34	67.50
	Purge-Gate (C1)	13.79	37.07	4.17	72.97
	PEA (C2)	13.83	36.90	4.27	72.85
	PEA-ML (C2)	11.16	31.88	3.87	74.46
CT	LN (B)	9.97	46.48	5.59	65.52
	PEA (C2)	13.98	36.77	4.20	72.18

Source only : PARE-Net. EP : episodic. CT: continual

Results : input adaptation is the most robust TTA family across datasets, although the gains remain limited. LN degrades registration on P2ILReg.

Computational complexity

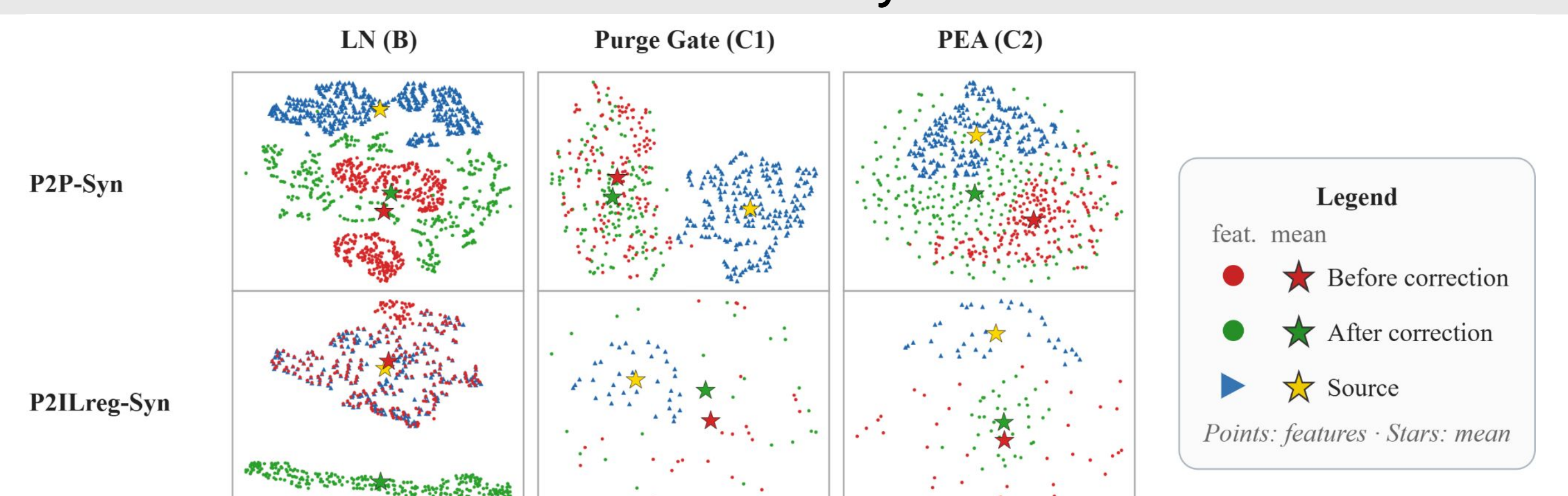


Ablation study : asymmetric vs symmetric adaptation

	Method	P2ILReg-Syn Syn → Corrupted syn		P2ILReg-Real Syn → Real			
		Symmetric	Asymmetric	Symmetric	Asymmetric		
		RMSE (mm)↓	RMSE (mm)↓	CD (mm)↓	Dice ↑	CD (mm)↓	Dice ↑
EP	Source only	38.70	38.70	4.26	71.87	4.26	71.87
	LN (B)	38.79	47.31	4.22	72.21	5.34	67.50
	Purge-Gate (C1)	37.39	37.07	4.34	71.51	4.17	72.97
	PEA (C2)	38.04	36.90	4.43	72.21	4.27	72.85

- Purge-Gate and PEA: asymmetric adaptation improves results on P2ILReg.
- LN: symmetric adaptation performs better.

Visual analysis



t-SNE visualization : only limited features realignment is observed.

Conclusion

- Input adaptation improves registration on all datasets, at low latency
- Normalization adaptation helps P2P only : unreliable target statistics
- Model adaptation : slower and more GPU memory

Acknowledgments

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